WHANHEE CHO

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RESEARCH INTERESTS

Data Summarization, Data Interpretability and Explanation, Applied LLMs

EDUCATION

UNIVERSITY OF UTAH, Graduate School, Utah, USA — Aug. 2023 - Present
Doctor of Philosophy in Computer Science

HANYANG UNIVERSITY, Seoul, Republic of Korea — Mar. 2021 - Feb. 2023
Master of Science in Computer Science
Summa Cum Laude

HANYANG UNIVERSITY, Seoul, Republic of Korea — Mar. 2017 - Feb. 2021
Bachelor of Computer Science and Software Engineering
Summa Cum Laude

PUBLICATIONS

- "QUWARTS: Query Workload Aware Relational Table Synthesis from Unstructured Text", *VLDB*, 2026
  Aritra Mazumder, **Whanhee Cho**, Anna Fariha
- "SAGE: Adaptive Recommendation of Spreadsheet Pivot Tables", *SIGMOD*, *2026*
  Whanhee Cho, Shamit Fatin, Anna Fariha
- "Data-Semantics-Aware Recommendation of Diverse Pivot Tables", *SIGMOD, 2026*
  Whanhee Cho, Anna Fariha
- "UTOPIA: Automatic Pivot Table Assistant", *PVLDB, 2024*
  Whanhee Cho, Anna Fariha
- "Simple Data Transformations for Mitigating the Syntactic Similarity to Improve Sentence Embeddings at Supervised Contrastive Learning", *Advanced Intelligent Systems, 2024*
  Minji Kim, **Whanhee Cho**, Soohyeong Kim, and Yong Suk Choi.
- "Bidirectional Masked Self-attention and N-gram Span Attention for Constituency Parsing", *EMNLP Findings, 2023*
  Soohyeong Kim, **Whanhee Cho**, Minji Kim, and Yong Suk Choi

PREPRINT

- "Example-Driven Intent Synthesis for Constrained Data Bundle Retrieval: Focused Text Snippet Extraction and Beyond", 2026
  Whanhee Cho, Kuangfei Long, Mahmood Jasim, Matteo Brucato, Alexandra Meliou, Peter J. Haas, Anna Fariha
- "CLUE: Answering How-To Questions on Dashboards using Metric Lineage Extractor and Simulator" 2026
  Wei-Xuan Wang, **Whanhee Cho**, Mahmood Jasim, Anna Fariha
- "ACCIO: Table Understanding Enhanced via Contrastive Learning with Aggregations.", 2024
  Whanhee Cho

DATABASE MANAGEMENT RESEARCH EXPERIENCE

University of Utah DBLAB — Salt Lake City, USA
Research Assistant — Aug. 2023 - Present

- Intent-Aligned Interactive Bundle Retrieval
  - Developed a novel algorithmic framework for detecting multi-dimensional user intents for bundle retrieval without explicit prompts, resolving combinatorial explosion via Integer Linear Programming (ILP) with PaQL. Demonstrated superior performance to RAGs on extractive summarization tasks.
- Diversified Interesting Query Result Recommendation (Accepted at SIGMOD 2026)
  - Designed a recommendation system that evaluates the usability of query results and leverages Large Language Models (LLMs) to capture human common sense, successfully increasing the surprisingness and utility of the output.
  - Solved the combinatorial explosion of candidate sets with pruning strategies, reducing candidate evaluation cost by 80% while maintaining quality. We evaluated our work and novel metrics via a formal user study.

- Semantic aware data post-processing in the pivot table (Published at VLDB 2024)
    - Designed a system that leverages language models and clustering algorithms to automatically group related entities and correct typos, delivering semantically accurate pivot tables in spreadsheet.
- Table Understanding with Contrastive Learning
    - Introduced novel contrastive learning paradigm using related table pairs, achieving 91.1% micro-F1 on colu mn type annotation. Demonstrated competitive performance approaching state-of-the-art methods (within 4 % on micro-F1) while using conceptually simpler approach without data augmentation.

AI RESEARCH EXPERIENCE

Hanyang University Artificial Intelligence LAB Seoul, South Korea
Full-time Student Researcher Dec. 2020 - Feb. 2023

- Proposed and implemented a method to syntactically augment a dataset to improve sentence embedding performance. Our work demonstrates that simple transformations can have a significant impact on model performance, addressing the limitations of existing semantic-only approaches
- Leverage the prior works' exploration that the internal layers of language encoders learn different aspects of a sentence. We demonstrated that using these varied outputs as input for a GAN produces a more meaningful representation than using random noise, which often fails in low-resource settings.
- Investigated relations and impacts of next token probability onto positional embeddings. The purpose of this project was to validate the effect of GPT-2 next token prediction probability on positional embedding at the relation extraction task using the TACRED dataset. According to the next token probability, set the positional embedding values of the next gold token to be closer or further from the current token.

TECHNICAL SKILLS

- Programming Languages: C/C++, Python, Java, SQL
- ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn, Huggingface
- Tools & Technologies: Git, CMake, Linux kernel programming, CUDA, Docker

CONFERENCE & PRESENTATION

- University of Utah, School of Computing, "Data Science Research," Salt Lake City, UT, USA, April 2025
- VLDB 2024, "UTOPIA: Automatic Pivot Table Assistant", Guangzhou, China, August 2024
- NWDS 2023, "UTOPIA: Automatic Pivot Table Assistant," Seattle, WA, USA, August 2023

TEACHING EXPERIENCE

Teaching Assistant | University of Utah

- CS6959/3960: Human-Centered Data Management | Spring 2026
- CS6530: Advanced Database Systems | Fall 2025

Teaching Assistant | Hanyang University

- ITE4053: Deep Learning Methods and Applications | Fall 2022
- CSE4007: Artificial Intelligence | Fall 2021
- CSE0092: Introduction to Artificial Intelligence | Fall 2021, Spring 2022

WORK EXPERIENCE

DAYONE COMPANY, SAMSUNG DS Seoul, South Korea
Lecturer April 2023 - June 2023

- Lectured about basic python programming, data preprocessing, and probability and statistics.

AWARDS & HONORS

Fellowship, Kahlert School of Computing, University of Utah	2023
Brain Korea 21 Fellowship, NRF Korea	2021-2022
Need Based Scholarship, Hanyang University	2019-2020
Merit Based Scholarship, Hanyang University	2018, 2021-2022
Honors student in the 1st semester of 2018, Hanyang University	2018
3rd Prize, University-wide Group Study Competition	2018
National Grant, Korea Student Aid Foundation (KOSAF)	2017- 2020